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Signal Processing

- Discrete Time Fourier Transform (DTFT)
- Examples
- DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n} \quad \text{(Analysis)}$$

$\omega \rightarrow$ frequency index / variable

$n \rightarrow$ time index

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\cdot e^{j\omega n}\, d\omega \quad \text{(Synthesis)}$$

- Examples:

$$h_{ave}[n] = \frac{\delta[n] + \delta[n-1]}{2} \;; \quad h_{diff}[n] = \frac{\delta[n] - \delta[n-1]}{2}$$

$$H_{ave}(e^{j\omega}) = \frac{1 + e^{-j\omega}}{2} \;; \quad H_{diff}(e^{j\omega}) = \frac{1 - e^{-j\omega}}{2}$$

$$H_{ave}(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_{ave}[n]\, e^{-j\omega n}$$

$$= \frac{1}{2}\cdot e^{-j\omega 0} + \frac{1}{2}\cdot e^{-j\omega}$$

$$|H_{ave}(e^{j\omega})| = \left| e^{-j\frac{\omega}{2}} \left(\cos\frac{\omega}{2}\right) \right|$$

$$= \left|\cos\frac{\omega}{2}\right| \Rightarrow$$